

University AI Assistant Project

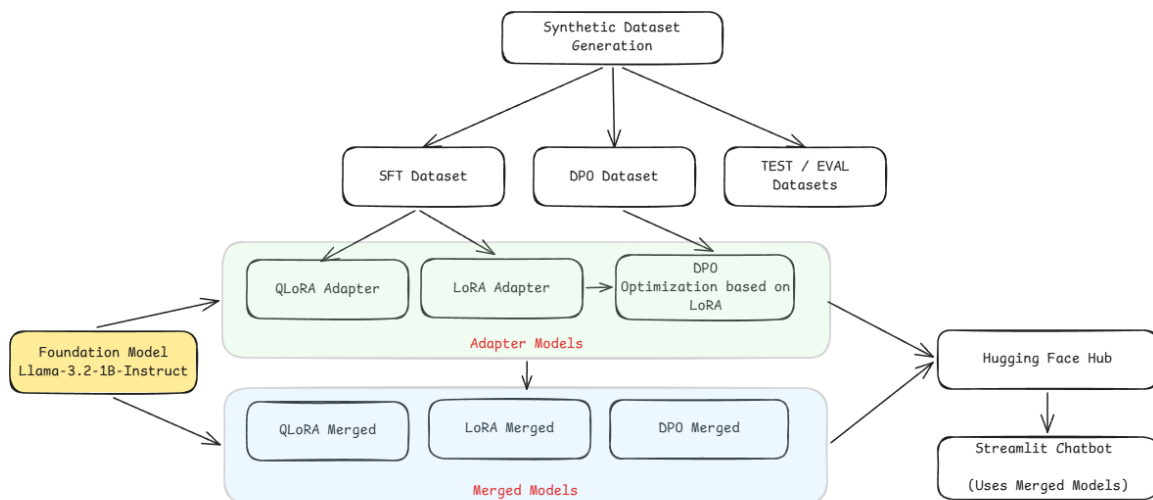
Project Overview

University AI Assistant is an end-to-end Generative AI project developed to demonstrate parameter-efficient fine-tuning of a Large Language Model for a university helpdesk use case. The project covers synthetic dataset generation, supervised fine-tuning using LoRA and QLoRA, preference optimization using Direct Preference Optimization (DPO), model evaluation, deployment to Hugging Face, and a Streamlit-based chatbot.

Project Objectives

- Develop a domain-specific AI assistant for university queries.
- Compare Foundation, LoRA, QLoRA and DPO models.
- Understand parameter-efficient fine-tuning techniques.
- Deploy the solution using Hugging Face and Streamlit.

Data/Model Flow



Project Implementation

- Fine-tuned Llama-3.2-1B-Instruct using LoRA, QLoRA and DPO.
- Created five synthetic datasets covering training, test and evaluation.

- Published merged deployment models to Hugging Face.
- Developed a Streamlit chatbot with runtime model selection.
- Compared Foundation, LoRA, QLoRA and DPO responses.

Engineering Decisions

- Kaggle was selected for GPU availability.
- Unsloth was chosen for faster fine-tuning.
- Synthetic datasets were created because suitable university datasets were unavailable.
- Adapter models were merged for simpler deployment.
- Hugging Face was used as the model registry.
- Streamlit provided a lightweight demonstration interface.

Technology Stack

Component	Technology
Development	Kaggle (Tesla T4 GPU)
Language	Python
Libraries	Unsloth, Transformers, PEFT, TRL, Datasets
Model Repository	Hugging Face
Deployment	Streamlit

Datasets

Five datasets were created specifically for the university domain. Two independent dataset versions were synthetically generated to train and evaluate the models, enabling a comparison of how dataset quality influences model performance and inference.

Dataset	Purpose
SFT Dataset	Prompt-response pairs used for supervised fine-tuning.
DPO Dataset	Chosen and rejected responses used for preference learning.
TEST Dataset	Questions with expected answers for qualitative validation.
EVAL Dataset	Independent evaluation questions with varying difficulty.

Models Developed

The project produced adapter models for training and merged models for deployment.

Model	Training	Deployment
Foundation	Pre-trained	Base model
LoRA	Adapter	Merged model
QLoRA	Adapter	Could not generate due to error
DPO	Adapter	Merged model

Links to Model

Model	Hugging Face Link
Foundation	meta-llama/Llama-3.2-1B-Instruct · Hugging Face
LoRA Adapter	https://huggingface.co/Akay2026/University-AI-LoRA
LoRA Merged	https://huggingface.co/Akay2026/University-AI-LoRA-Merged
qLoRA Adapter	https://huggingface.co/Akay2026/University-AI-QLoRA
qLoRA Merged	Could not generate the merged QLoRA model due to a Transformers compatibility issue .
DPO Adapter	https://huggingface.co/Akay2026/University-AI-DPO
DPO Merged	https://huggingface.co/Akay2026/University-AI-DPO-Merged

Deliverables

1. Datasets

- Version 1: Initial synthetic dataset containing brief responses. This dataset was used for the initial model training and experimentation.
- Version 2: Enhanced synthetic dataset with richer and more detailed responses. The current LoRA, QLoRA, and DPO models were trained using this dataset.

2. Streamlit Application

- app.py – A lightweight chatbot application that enables users to interact with the Foundation, LoRA, QLoRA, and DPO models through a simple web interface.

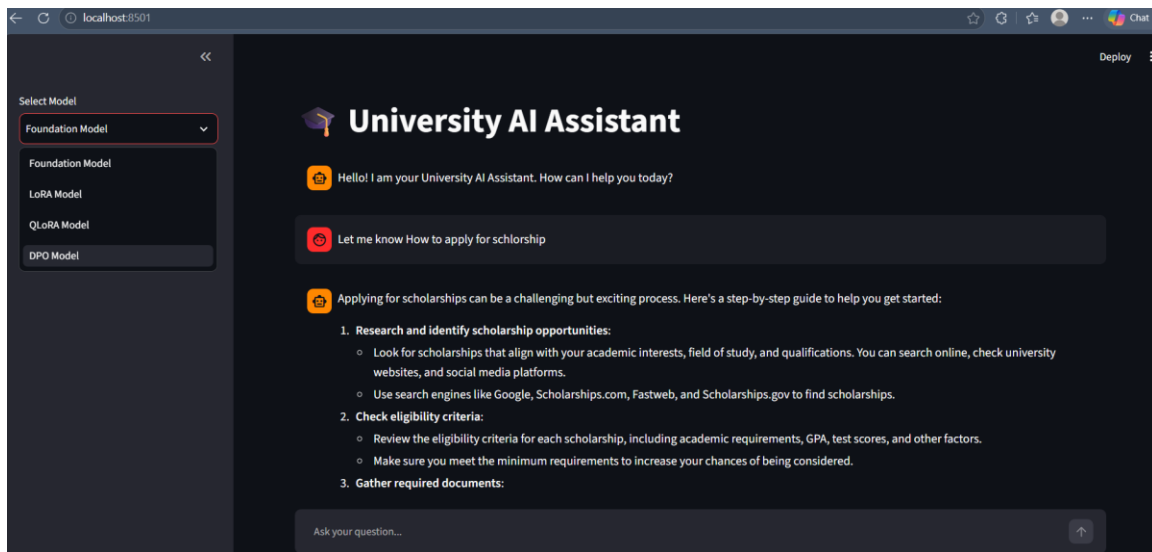
3. Training Notebook

- University-AI-Assistant-Finetuning.ipynb – Contains the complete implementation for dataset preparation, model loading, fine-tuning (LoRA, QLoRA, and DPO), model evaluation, model merging, and uploading the trained models to Hugging Face.

Evaluation and Streamlit

The Foundation, LoRA, QLoRA and DPO models were compared using a common set of university-domain questions. Responses were evaluated qualitatively for relevance, correctness and completeness.

A lightweight Streamlit chatbot allows users to choose a model, submit questions and compare the generated responses while maintaining conversation history.



Results and Observations

The Foundation model produced fluent but generic responses. The performance of the LoRA, QLoRA, and DPO models was largely influenced by the quality of the training dataset, making it difficult to draw definitive conclusions about the fine-tuning techniques. Models trained on a lightweight dataset generated short responses and occasionally failed on out-of-domain queries, while models trained on a richer dataset produced more detailed but sometimes less relevant answers. LoRA adapted well to university-specific terminology, QLoRA achieved similar performance with lower memory requirements, and DPO showed improved response alignment when supported by a high-quality preference dataset. Merged models also simplified deployment in the Streamlit application.

Challenges Faced

- Library version mismatches across Unsloth, Transformers and TRL.
- Unable to generate the merged QLoRA model due to a compatibility issue between Unsloth and the Transformers library during model serialization (weight reversal).
- Preparing high-quality SFT and DPO datasets.
- Generating meaningful chosen/rejected preference pairs.
- Tokenizer compatibility while deploying merged models.
- GPU dependency of Unsloth for local inference.
- Hugging Face download and caching issues during development.

Future Improvements

- Improve dataset quality and diversity. Use publicly available rich data sources
- Adopt an automated evaluation framework such as DeepEval.

- Use Jinja chat templates.
- Experiment with GRPO and newer alignment techniques.
- Tune training hyperparameters.
- Generate and deploy models in GGUF format for local inference using tools such as Ollama.
- Integrate Retrieval-Augmented Generation (RAG) using university documents.

8. Conclusion

The project successfully demonstrated the complete lifecycle of adapting a foundation language model to a university-specific domain using modern parameter-efficient fine-tuning techniques. The resulting models were evaluated, deployed on Hugging Face, and integrated into a Streamlit chatbot, providing a practical end-to-end Generative AI solution while highlighting real-world engineering challenges and lessons learned.

Annex 1 : Screenshots of Training Log of while training the models

LORA

Unsloth: Double buffering enabled (parallel H2D + compute) for backward pass.

 [125/125 04:50, Epoch 1/1]

Step	Training Loss
------	---------------

10	3.685995
----	----------

20	2.795690
----	----------

30	1.670321
----	----------

40	0.630091
----	----------

50	0.186190
----	----------

60	0.096109
----	----------

70	0.077392
----	----------

80	0.066674
----	----------

90	0.062806
----	----------

100	0.060936
-----	----------

110	0.058077
-----	----------

120	0.057843
-----	----------

Unsloth: Restored added_tokens_decoder metadata in ./outputs_sft/checkpoint-125/tokenizer_config.json.

Unsloth: Restored added_tokens_decoder metadata in models/University-AI-LoRA/tokenizer_config.json.

QLORA

 [125/125 05:35, Epoch 1/1]

Step	Training Loss
------	---------------

10	3.704221
----	----------

20	2.778035
----	----------

30	1.637127
----	----------

40	0.600798
----	----------

50	0.173333
----	----------

60	0.093630
----	----------

70	0.075281
----	----------

80	0.065927
----	----------

90	0.062480
----	----------

100	0.060509
-----	----------

110	0.057487
-----	----------

120	0.057391
-----	----------

Unsloth: Restored added_tokens_decoder metadata in models/University-AI-QLoRA/tokenizer_config.json.

DPO

Unsloth: Will smartly offload gradients to save VRAM!
Unsloth: Double buffering enabled (parallel H2D + compute) for backward pass.

[125/125 12:37, Epoch 1/1]

Step	Training Loss	rewards / chosen	rewards / rejected	rewards / accuracies	rewards / margins	logps / chosen	logps / rejected	logits / chosen	logits / rejected
10	0.561490	0.004605	-0.472491	0.300000	0.477096	-79.328529	-1827.605225	-1.428782	-2.137747
20	0.001204	0.132024	-17.903181	1.000000	18.035206	-79.696915	-1991.699463	-1.219172	-2.123928
30	0.000000	0.153157	-42.599712	1.000000	42.752872	-71.194313	-2203.916260	-0.978251	-2.006858
40	0.000000	0.046717	-54.013416	1.000000	54.060131	-76.704147	-2365.084717	-0.838470	-1.917616
50	0.000000	-0.109427	-56.716469	1.000000	56.607044	-77.590691	-2395.544434	-0.796553	-1.854321
60	0.000000	-0.161444	-56.972736	1.000000	56.811291	-72.683487	-2388.527344	-0.818611	-1.823712
70	0.000000	-0.087204	-58.337639	1.000000	58.250446	-76.665184	-2373.433105	-0.788101	-1.850354

Annex 2 : Benchmark

	id	difficulty	question	Foundation Response	LoRA Response	QLoRA Response	DPO Response	Foundation Length	LoRA Length	QLoRA Length	DPO Length
0	EVAL-0001	Easy	Who is the primary point of contact or office ...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	62	86	194	68
1	EVAL-0002	Medium	If my initial application for a policy adjustm...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	197	211	126	106
2	EVAL-0003	Hard	Can you outline the comprehensive strategy for...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	system\n\nCutting Knowledge Date: December 202...	194	209	211	209

Benchmark Summary

Evaluation Questions : 3

Average Response Length (Words)

Foundation Model : 151.0

LoRA Model : 168.7

QLoRA Model : 177.0

DPO Model : 127.7

Longest Response

Foundation : 197 words

LoRA : 211 words

QLoRA : 211 words

DPO : 209 words

Shortest Response

Foundation : 62 words

LoRA : 86 words

QLoRA : 126 words

DPO : 68 words

Generated Files

output/Benchmark_Comparison.xlsx